



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to PFSCS LTD

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

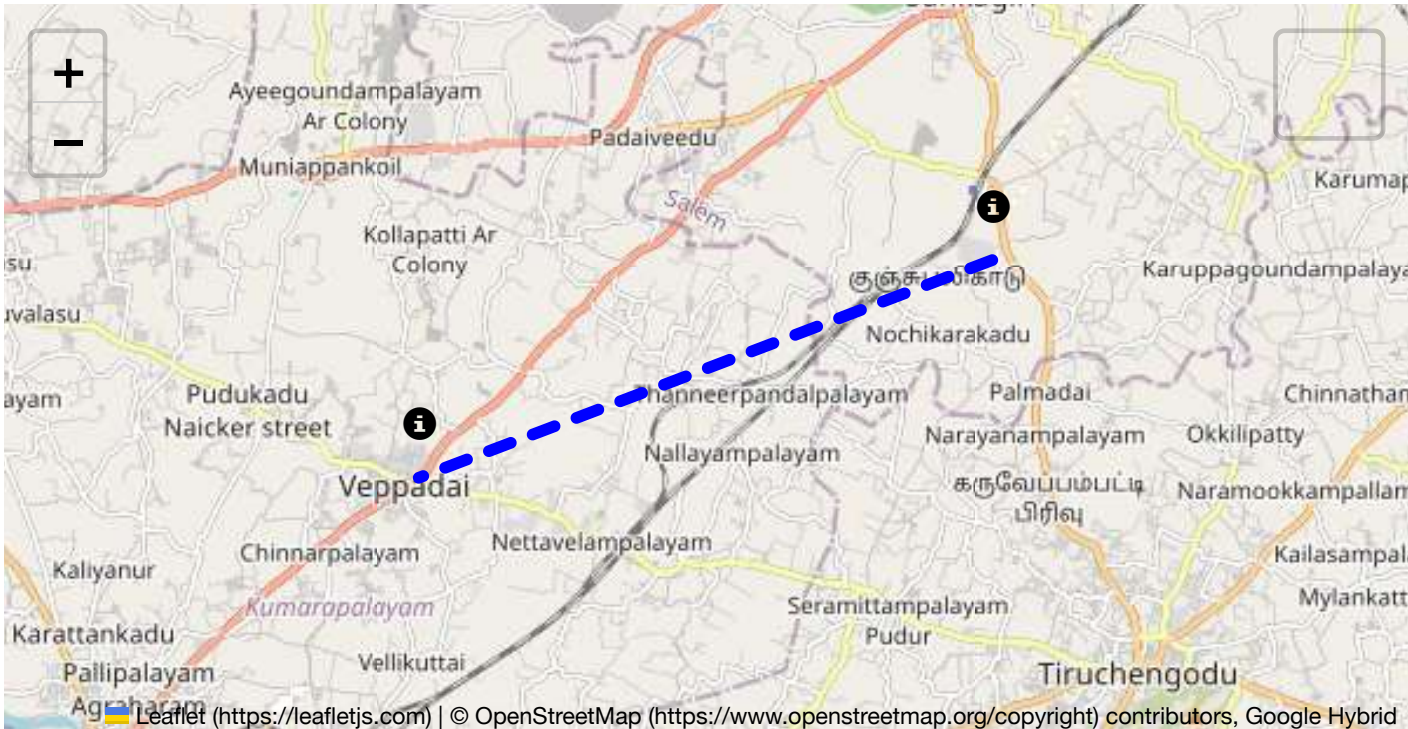
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

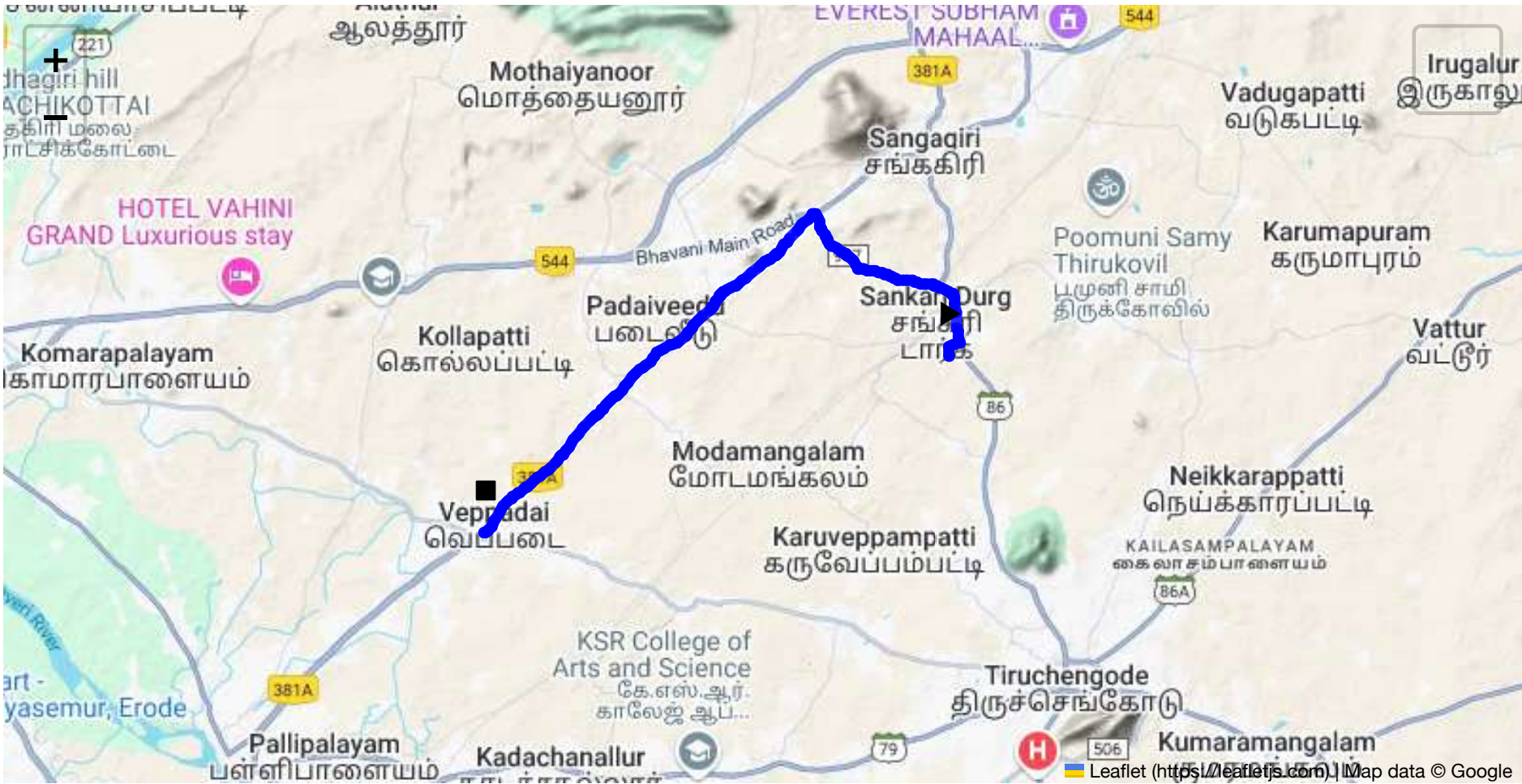
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 13.98 km
Estimated Duration: 0.3 hours
Adjusted Duration (Heavy Vehicle): 0.4 hours

Start: (11.4381, 77.8734)
End: (11.406584, 77.789494)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to MGR Nagar, Veppadai, Tamil Nadu, spans approximately 13.98 kilometers. This relatively short journey typically takes around 20 minutes for heavy vehicles under normal traffic conditions. The path primarily runs along rural and semi-urban roads, connecting two important local economic centers.

Typical Weather Conditions and Potential Hazards

The region experiences a tropical climate, with the following seasonal variations:

- **Summer (March-May):** High temperatures can lead to vehicle overheating.
- **Monsoon (June-September):** Heavy rainfall may cause road flooding and reduced visibility, increasing the likelihood of accidents.
- **Post-Monsoon (October-November):** Occasional thunderstorms can disrupt travel.
- **Winter (December-February):** Generally mild but with occasional fog, which can affect visibility.

Truck drivers should be prepared for reduced traction and slower travel during the monsoon season due to wet road conditions.

Traffic Patterns

- **Peak Hours:** Morning (8 AM to 10 AM) and evening (5 PM to 7 PM) experience the highest traffic volumes due to local commuting patterns.
- **Congestion-Prone Areas:** Small towns and junctions along the route, particularly during market days, can experience increased congestion.

Road Quality and Infrastructure

The road condition varies from well-maintained highways to narrower, less-paved rural roads:

- **Highway Sections:** Good quality with clear markings and signage.
- **Rural Roads:** Can be uneven and may lack proper markings and illumination.

Alternative Routes for Emergencies

In the event of disruptions along the primary route:

- **Via NH44:** Offers a slightly longer but more robust infrastructure, suitable for navigating around potential roadblocks or floods.
- **Local Road Detours:** Knowing local routes or consulting with logistic coordinators can provide quick detours around isolated incidents.

Local Regulations Affecting Hazardous Material Transport

- **Permits Required:** Transport of hazardous materials requires special permits and adherence to prescribed safety regulations.
- **Restricted Hours:** Certain areas might have restrictions on vehicle types during school hours or other community events.

Historical Incidents Involving Heavy Vehicles or Hazardous Materials

While no major incidents are recorded over the past few years for this specific route, general causes of heavy vehicle accidents in the region include:

- Overloading
- Brake failure on steep inclines
- Poor weather-related visibility

Environmental Considerations and Sensitive Areas

- **Agricultural Zones:** The route passes through areas with significant agricultural activities where spills could cause significant environmental damage.
- **Wildlife Crossings:** Look out for areas marked for animal crossings, particularly during dawn and dusk.

Communication Coverage

- **Strong Coverage:** Major towns and along the highways.
- **Potential Dead Zones:** Remote sections of rural roads may experience weak or no mobile signal, limiting communication in emergencies.

Estimated Emergency Response Times

- **Urban Areas:** Roughly 15-30 minutes due to proximity to local emergency services.
- **Rural Areas:** Could extend to 45 minutes or more due to access issues and distance.

Overall Risk Assessment Summary

The journey from Sangagiri to MGR Nagar, Veppadai involves moderate risk levels primarily due to weather conditions, road quality, and traffic congestion. While well-maintained highway sections ease movement, poorer rural road conditions and potential monsoon impacts require careful attention. Planning for communication barriers and adjusting travel during peak times or inclement weather conditions can mitigate potential risks. Transport companies should ensure that drivers are well-trained in emergency procedures and communication is maintained through reliable channels to enhance safety along the route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.46307, 77.84941	30 KM/Hr
5	Turn	Medium	11.46318, 77.84913	30 KM/Hr
6	Turn	Medium	11.46312, 77.84886	30 KM/Hr
7	Blind Spot	Blind Spot	11.40664, 77.78942	10 KM/Hr

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46318, 77.84913



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46312, 77.84886



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.40664, 77.78942